

$$\min_x f(x) = \min_x (f^1(x) + f^2(x))$$

$$\begin{aligned}\min_x f(x) &= \min_x (f^1(x) + f^2(x)) \\ &= \min_{x^1=x^2} (f^1(x^1) + f^2(x^2))\end{aligned}$$

$$\begin{aligned}\min_x f(x) &= \min_x (f^1(x) + f^2(x)) \\ &= \min_{x^1=x^2} (f^1(x^1) + f^2(x^2)) \\ &= \min_{x^1, x^2} (f^1(x^1) + f^2(x^2) + \mathbb{I}_\infty(x^1 = x^2))\end{aligned}$$

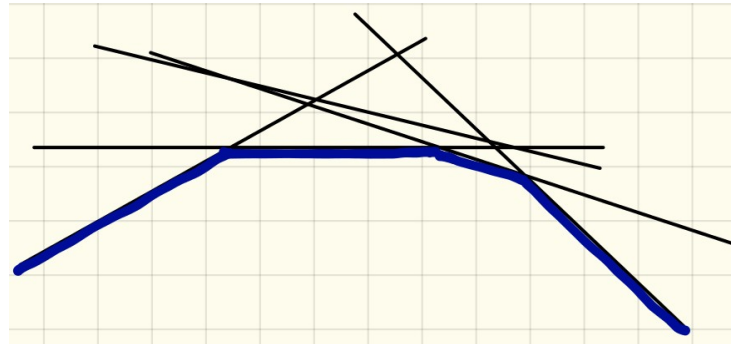
$$\begin{aligned}\min_x f(x) &= \min_x (f^1(x) + f^2(x)) \\ &= \min_{x^1=x^2} (f^1(x^1) + f^2(x^2)) \\ &= \min_{x^1, x^2} (f^1(x^1) + f^2(x^2) + \mathbb{I}_\infty(x^1 - x^2)) \\ &\geq \min_{x^1, x^2} (f^1(x^1) + f^2(x^2) + \lambda(x^1 - x^2))\end{aligned}$$

Lagrangian (Dual) Decomposition

$$\begin{aligned}\min_x f(x) &= \min_x (f^1(x) + f^2(x)) \\&= \min_{x^1=x^2} (f^1(x^1) + f^2(x^2)) \\&= \min_{x^1, x^2} (f^1(x^1) + f^2(x^2) + \mathbb{I}_\infty(x^1 - x^2)) \\&\geq \min_{x^1, x^2} (f^1(x^1) + f^2(x^2) + \lambda(x^1 - x^2)) \\&= \min_{x^1} (f^1(x^1) + \lambda x^1) + \min_{x^2} (f^2(x^2) - \lambda x^2) = u(\lambda)\end{aligned}$$

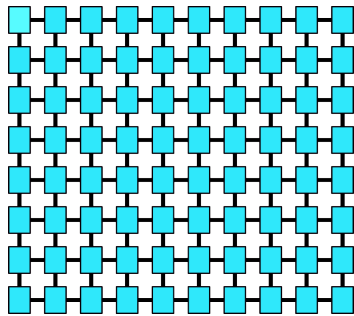
Lagrangian (Dual) Decomposition

$$\begin{aligned}\min_x f(x) &= \min_x (f^1(x) + f^2(x)) \\&= \min_{x^1=x^2} (f^1(x^1) + f^2(x^2)) \\&= \min_{x^1, x^2} (f^1(x^1) + f^2(x^2) + \mathbb{I}_\infty(x^1 = x^2)) \\&\geq \min_{x^1, x^2} (f^1(x^1) + f^2(x^2) + \lambda(x^1 - x^2)) \\&= \min_{x^1} (f^1(x^1) + \lambda x^1) + \min_{x^2} (f^2(x^2) - \lambda x^2) = u(\lambda)\end{aligned}$$



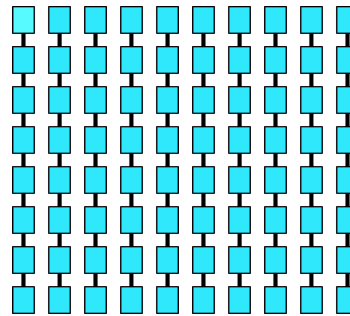
$u(\lambda)$ – concave

Lagrangian (Dual) Decomposition



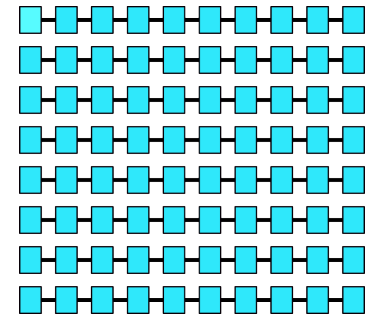
$$\langle \theta, \delta(x) \rangle$$

=

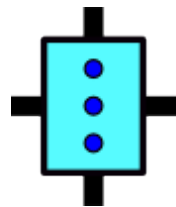


$$\langle \theta^c, \delta(x) \rangle$$

+

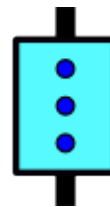


$$\langle \theta^r, \delta(x) \rangle$$

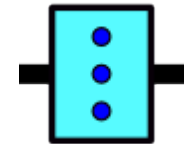


$$\theta_v(x_v)$$

=



+



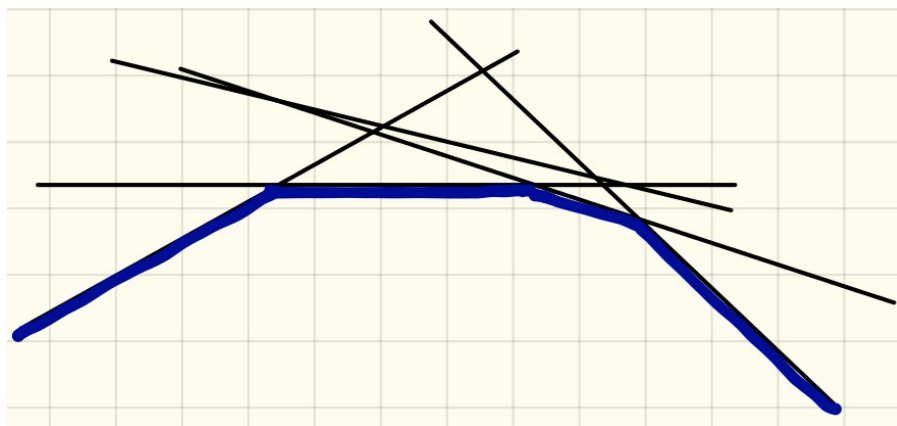
$$\frac{\theta_v(x_v)}{2} + \lambda_v(x_v) \quad + \quad \frac{\theta_v(x_v)}{2} - \lambda_v(x_v)$$

$$\min_{x \in \mathcal{X}} \langle \theta, \delta(x) \rangle \quad \stackrel{\forall \lambda}{\geq} \quad \min_{x^c \in \mathcal{X}^c} \langle \theta_{[\lambda]}^c, \delta(x^c) \rangle + \min_{x^r \in \mathcal{X}^r} \langle \theta_{[\lambda]}^r, \delta(x^r) \rangle$$

Lagrangian (Dual) Decomposition

$$\begin{aligned} U(\lambda) &= \left[\min_{x^c \in \mathcal{X}^c} \left\langle \theta_{[\lambda]}^c, \delta(x^c) \right\rangle + \min_{x^r \in \mathcal{X}^r} \left\langle \theta_{[\lambda]}^r, \delta(x^r) \right\rangle \right] \\ &= \left[\min_{\mu^c \in \mathcal{L}^c} \left\langle \theta_{[\lambda]}^c, \mu^c \right\rangle + \min_{\mu^r \in \mathcal{L}^r} \left\langle \theta_{[\lambda]}^r, \mu^r \right\rangle \right] \end{aligned}$$

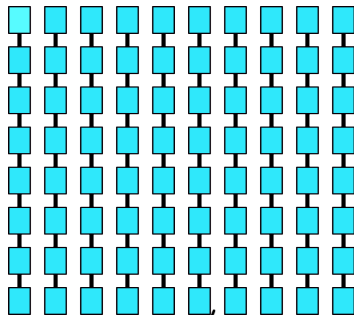
$$\theta_{[\lambda],v}^c = \frac{\theta_v}{2} + \lambda_v \quad \theta_{[\lambda],v}^r = \frac{\theta_v}{2} - \lambda_v \quad \theta_{[\lambda],uv}^c = \theta_{uv} \quad \theta_{[\lambda],uv}^r = \theta_{uv}$$



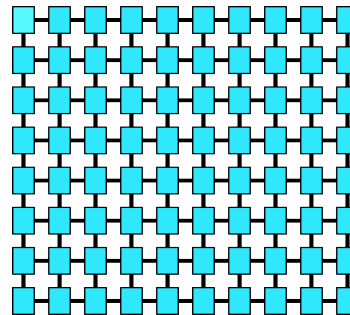
U is piece-wise linear, concave

Proposition: $\mu^* = \arg \min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle$

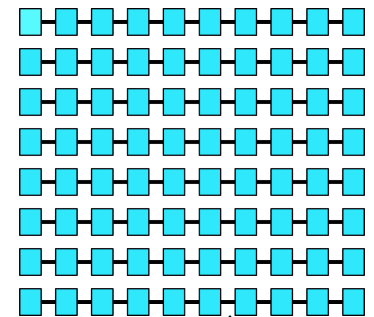
$$\left(\frac{\partial U}{\partial \lambda_v} = \mu_v^{c*} - \mu_v^{r*} = 0 \right) \Rightarrow \left(\mu_v^{c*} = \mu_v^{r*} \right)$$



$$\mu^{c*} = \min_{\mu^c \in \mathcal{L}^c} \langle \theta_{[\lambda]}^c, \mu^c \rangle$$



$$\mu^* = (\mu_v^c, \mu_{uv}^c, \mu_{uv}^r)$$

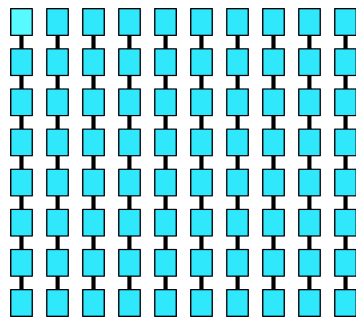


$$\mu^{r*} = \min_{\mu^r \in \mathcal{L}^r} \langle \theta_{[\lambda]}^r, \mu^r \rangle$$

Proposition:

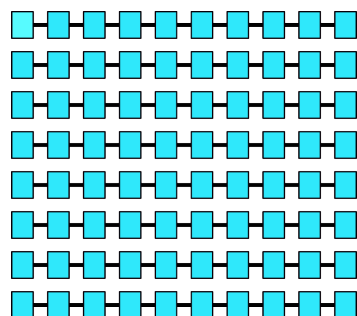
$$\left(\exists x^{c*}, x^{r*} : x^{c*} = x^{r*} = x^* \right) \Rightarrow \left(x^* = \arg \min_{x \in \mathcal{X}_v} \langle \theta, \delta(x) \rangle \right)$$

Equal slave integer labelings = Optimal integer labeling in the master problem


$$x^{c*} = \min_{x \in \mathcal{X}^c} \langle \theta_{[\lambda]}^c, \delta(x) \rangle$$

Column-wise labeling

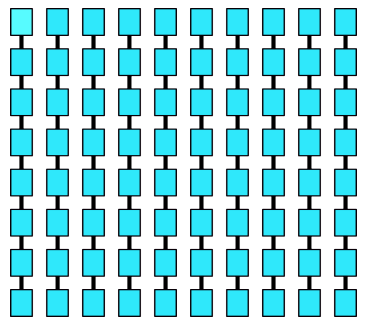
← Slave problems/solutions →


$$x^{r*} = \min_{x \in \mathcal{X}^r} \langle \theta_{[\lambda]}^r, \delta(x) \rangle$$

Row-wise labeling

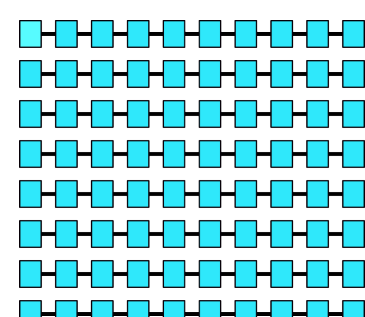
Obtaining approximate integer solution:

$$\hat{x} := \arg \min_{x^{c*}, x^{r*}} \{E(x^{c*}), E(x^{r*})\}$$


$$x^{c*} = \min_{x \in \mathcal{X}^c} \langle \theta_{[\lambda]}^c, \delta(x) \rangle$$

Column-wise labeling

← Slave problems/solutions →

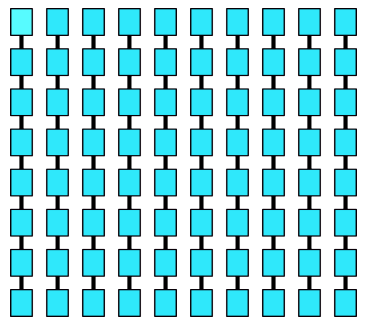

$$x^{r*} = \min_{x \in \mathcal{X}^r} \langle \theta_{[\lambda]}^r, \delta(x) \rangle$$

Row-wise labeling

Super-gradient:
$$\frac{\partial U}{\partial \lambda_v} = \delta(x^{c*})_v - \delta(x^{r*})_v$$

Sub-gradient method:

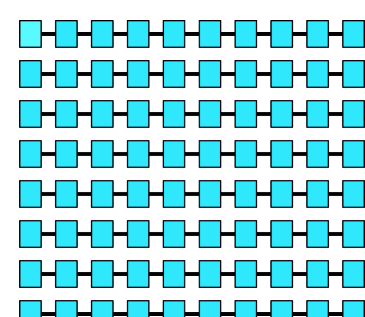
$$\lambda_v^{t+1} = \lambda_v^t + \alpha^t \frac{\partial U}{\partial \lambda_v} \quad \longrightarrow \quad \begin{aligned} \lambda_v^{t+1} &= \lambda_v^t + \alpha^t \delta(x^c)_v \\ \lambda_v^{t+1} &= \lambda_v^t - \alpha^t \delta(x^r)_v \end{aligned}$$



$$x^{c*} = \min_{x \in \mathcal{X}^c} \langle \theta_{[\lambda]}^c, \delta(x) \rangle$$

Column-wise labeling

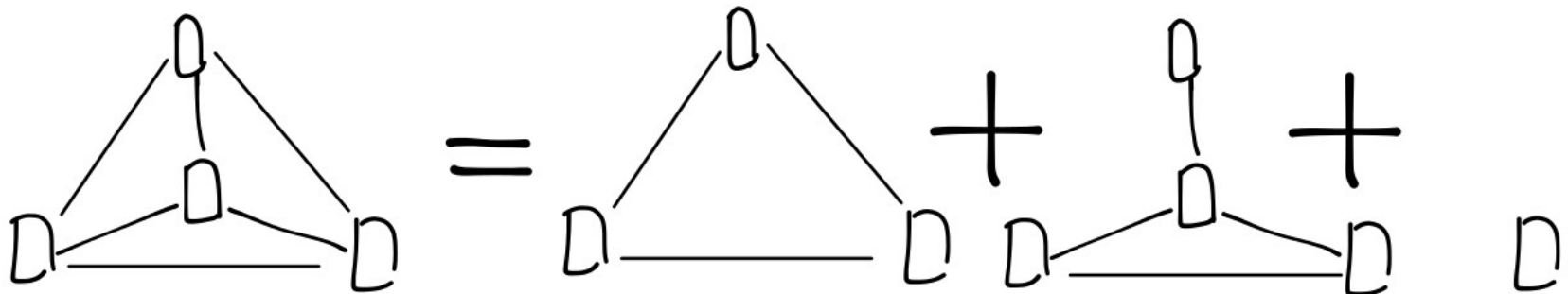
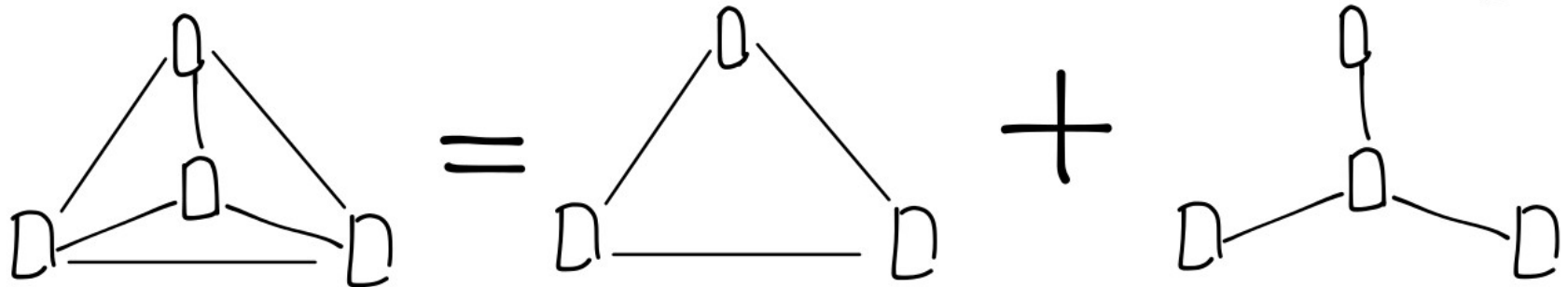
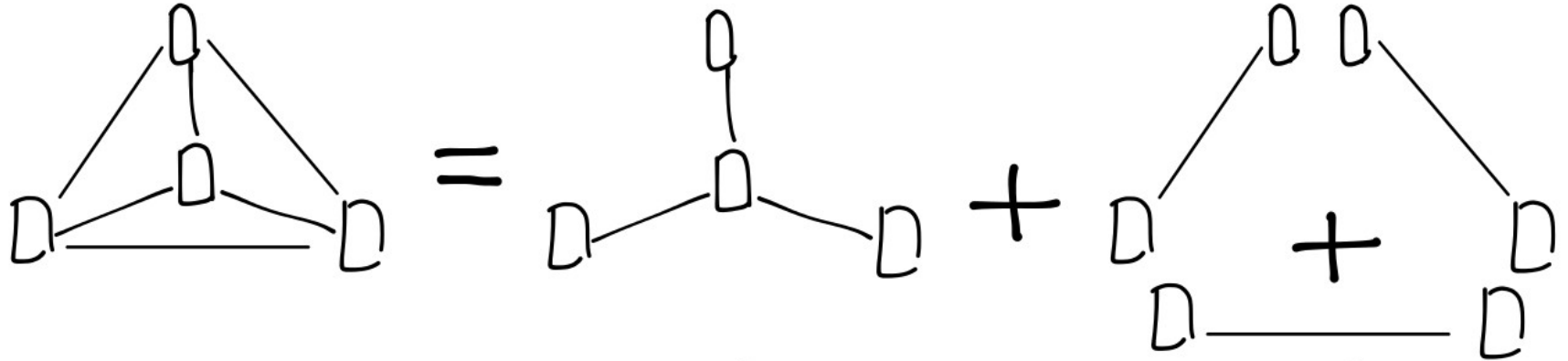
← Slave problems/solutions →



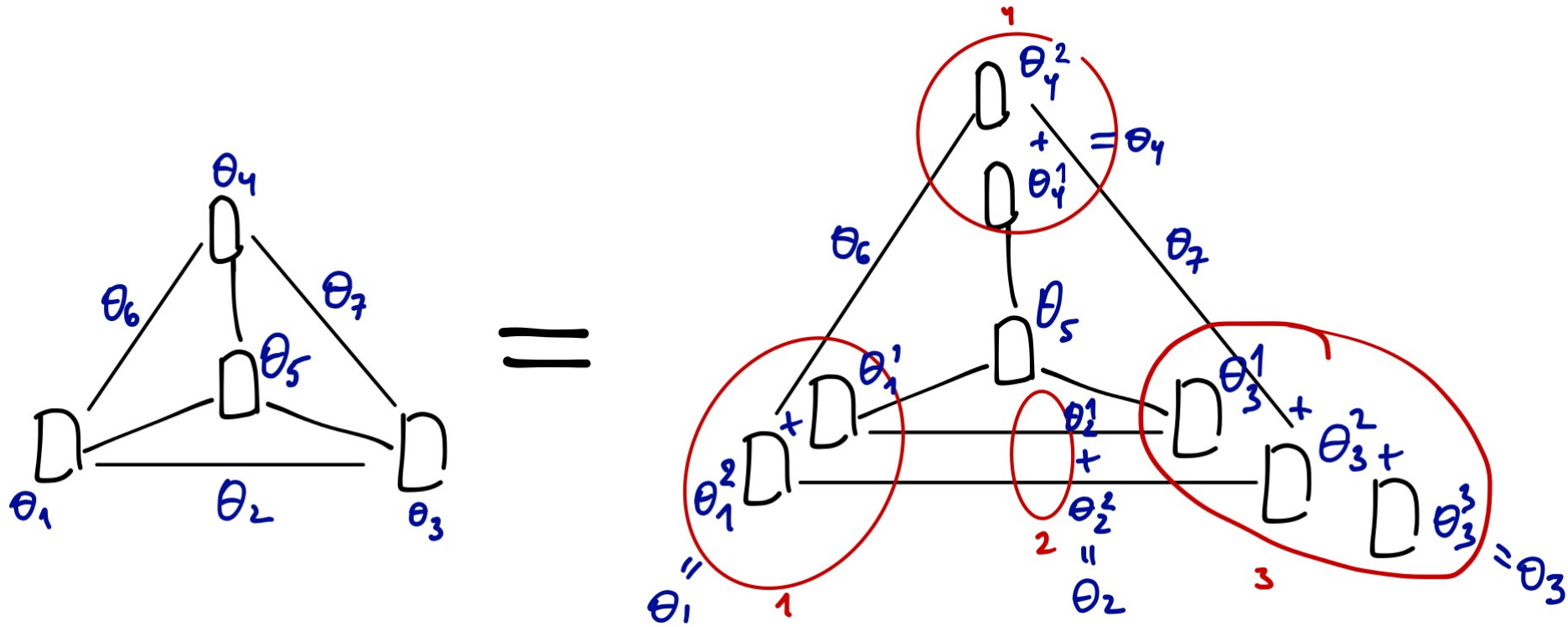
$$x^{r*} = \min_{x \in \mathcal{X}^r} \langle \theta_{[\lambda]}^r, \delta(x) \rangle$$

Row-wise labeling

General Sub-Graph-Wise Lagrangian Decomposition



General Sub-Graph-Wise Lagrangian Decomposition



$$\begin{aligned} \min_{x \in \mathcal{X}_v} E(x) &= \min_{x \in \mathcal{X}_v} \langle \theta, \delta(x) \rangle = \min_{x \in \mathcal{X}_v} \left\langle \sum_{\tau \in \mathcal{T}} \theta^\tau, \delta(x) \right\rangle \\ &\geq \sum_{\tau \in \mathcal{T}} \min_{x \in \mathcal{X}_{v^\tau}} \langle \theta^\tau, \delta(x) \rangle = \sum_{\tau \in \mathcal{T}} \min_{\mu \in \mathcal{M}^\tau} \langle \theta^\tau, \mu \rangle =: U(\theta^\mathcal{T}) \end{aligned}$$

$$\max_{\theta^{\mathcal{T}}} \sum_{\tau \in \mathcal{T}} \min_{x \in \mathcal{X}_{\mathcal{V}^{\tau}}} \langle \theta^{\tau}, \delta(x) \rangle = \max_{\theta^{\mathcal{T}}} \sum_{\tau \in \mathcal{T}} \min_{\mu \in \mathcal{M}^{\tau}} \langle \theta^{\tau}, \mu \rangle \quad \text{s.t.} \quad \sum_{\tau \in \mathcal{T}} \theta^{\tau} = \theta$$

Constrained optimization problem

Is the objective:

- 1) convex as a sum of minima of linear functions
- 2) concave as a sum of minima of linear functions
- 3) neither convex nor concave in general
- 4) neither of 1-2-3
- 5) no idea

$$\max_{\theta^\tau} \sum_{\tau \in \mathcal{T}} \min_{x \in \mathcal{X}_{\nu^\tau}} \langle \theta^\tau, \delta(x) \rangle = \max_{\theta^\tau} \sum_{\tau \in \mathcal{T}} \min_{\mu \in \mathcal{M}^\tau} \langle \theta^\tau, \mu \rangle \quad \text{s.t.} \quad \sum_{\tau \in \mathcal{T}} \theta^\tau = \theta$$

Constrained optimization problem

Do the constraints define:

- 1) a convex set as any equality constraints
- 2) a convex set as any linear equality constraints
- 3) not a convex set in general
- 4) neither of 1-2-3
- 5) no idea

$$\max_{\theta^{\mathcal{T}}} \sum_{\tau \in \mathcal{T}} \min_{x \in \mathcal{X}_{\mathcal{V}^{\tau}}} \langle \theta^{\tau}, \delta(x) \rangle = \max_{\theta^{\mathcal{T}}} \sum_{\tau \in \mathcal{T}} \min_{\mu \in \mathcal{M}^{\tau}} \langle \theta^{\tau}, \mu \rangle \quad \text{s.t.} \quad \sum_{\tau \in \mathcal{T}} \theta^{\tau} = \theta$$

Constrained optimization problem

Do the constraints define:

- 1) a bounded set as any equality constraints
- 2) a bounded set as any linear equality constraints
- 3) a linear subspace, therefore unbounded
- 4) neither of 1-2-3
- 5) no idea

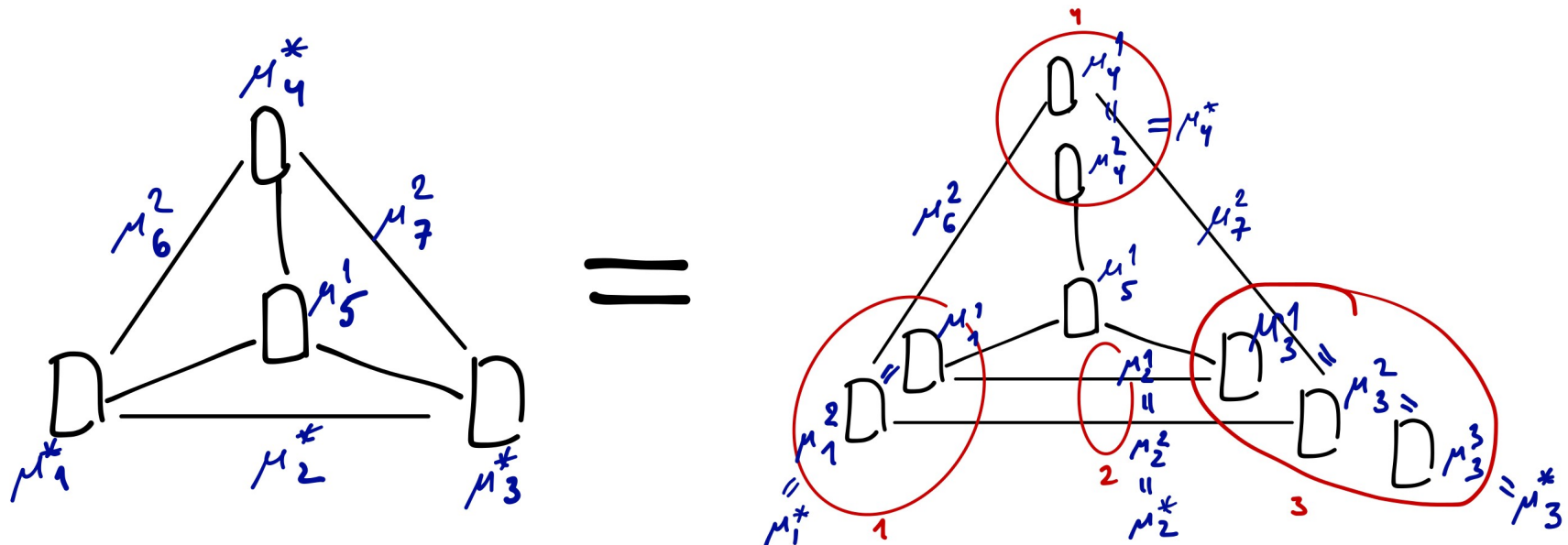
Dual Optimality Condition

$$\max_{\theta^\tau} \sum_{\tau \in \mathcal{T}} \min_{x \in \mathcal{X}_{v^\tau}} \langle \theta^\tau, \delta(x) \rangle = \max_{\theta^\tau} \sum_{\tau \in \mathcal{T}} \min_{\mu \in \mathcal{M}^\tau} \langle \theta^\tau, \mu \rangle \quad \text{s.t.} \quad \sum_{\tau \in \mathcal{T}} \theta^\tau = \theta$$

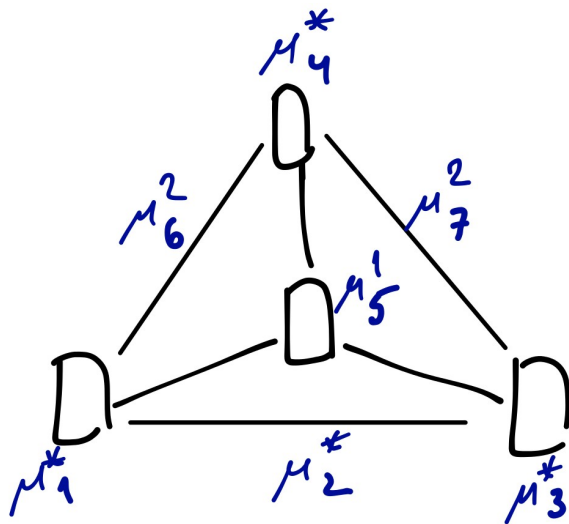
Constrained concave optimization problem over a linear subspace

$$\mu^\tau \in \arg \min_{\mu \in \mathcal{M}^\tau} \langle \theta^\tau, \mu \rangle$$

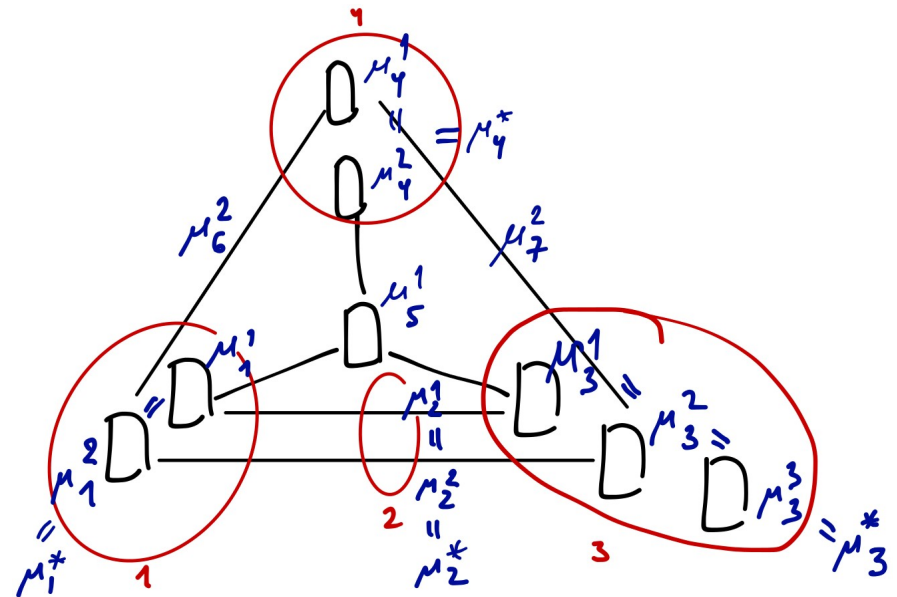
Relaxed optimality condition: $\exists \mu_w^\tau = \mu_w^\sigma = \mu_w^*$



Dual Optimality Condition



=



Exercise (may be difficult):

Formulate the respective primal problem, minimized by μ^*

$$\mu^* = \arg \min_{\mu \in ?} \langle \theta, \mu \rangle$$

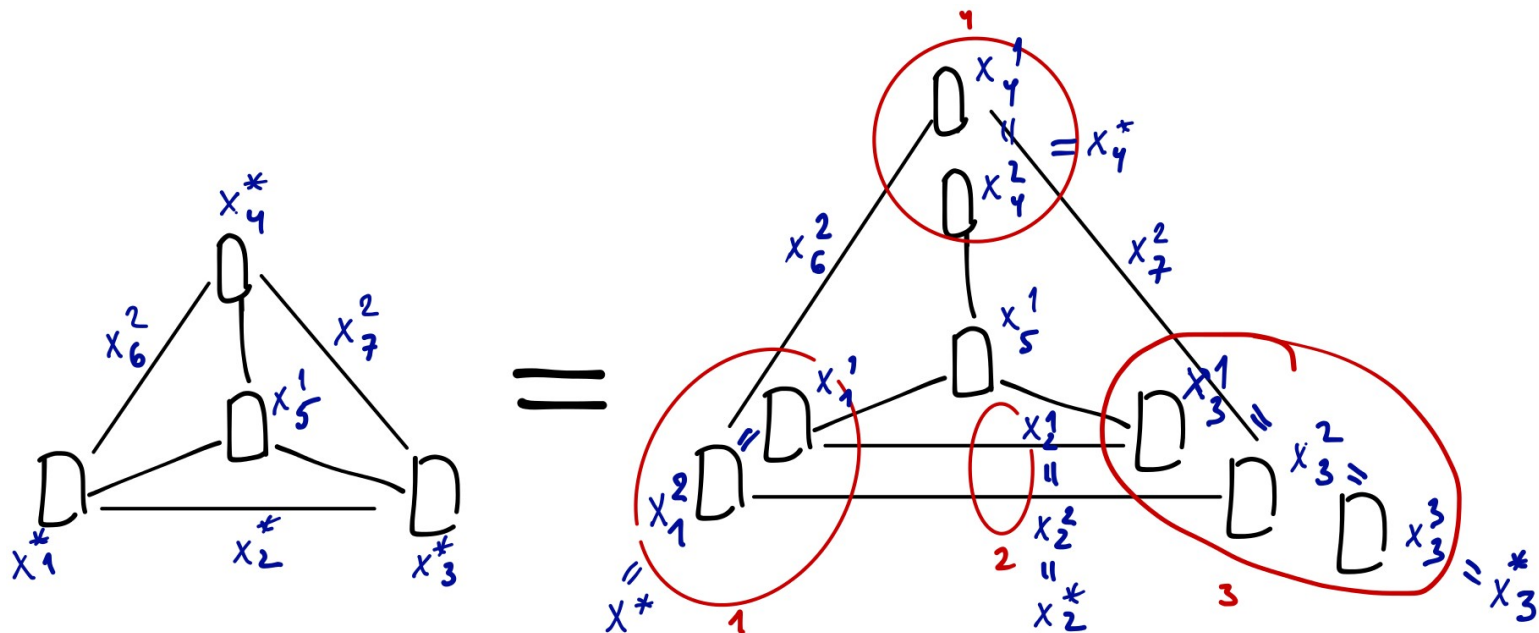
Non-Relaxed Optimality Condition

$$\max_{\theta^\tau} \sum_{\tau \in \mathcal{T}} \min_{x \in \mathcal{X}_{\nu^\tau}} \langle \theta^\tau, \delta(x) \rangle = \max_{\theta^\tau} \sum_{\tau \in \mathcal{T}} \min_{\mu \in \mathcal{M}^\tau} \langle \theta^\tau, \mu \rangle \quad \text{s.t.} \quad \sum_{\tau \in \mathcal{T}} \theta^\tau = \theta$$

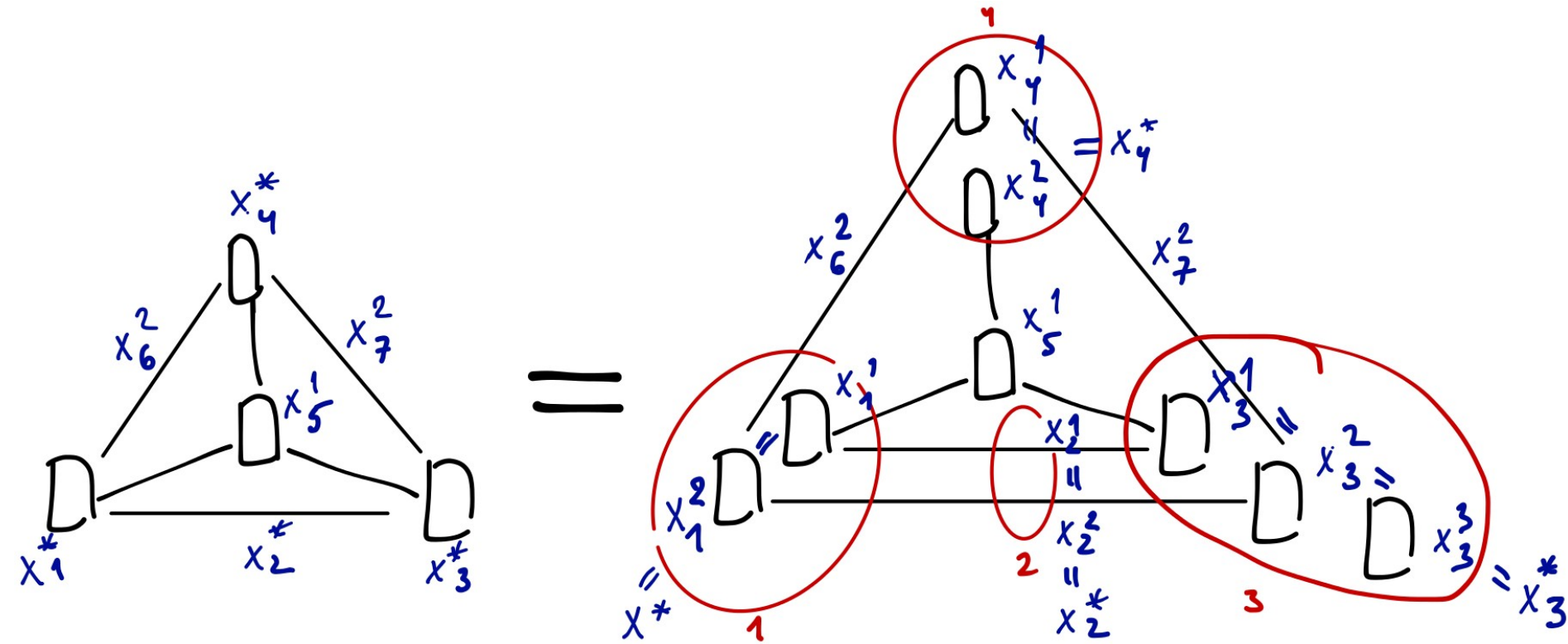
Constrained concave optimization problem over a linear subspace

$$x^\tau \in \arg \min_{x \in \mathcal{X}_{\nu^\tau}} \langle \theta^\tau, \delta(x) \rangle$$

Non-relaxed optimality condition: $\exists x_w^\tau = x_w^\sigma = x_w^*$



Non-Relaxed Optimality Condition

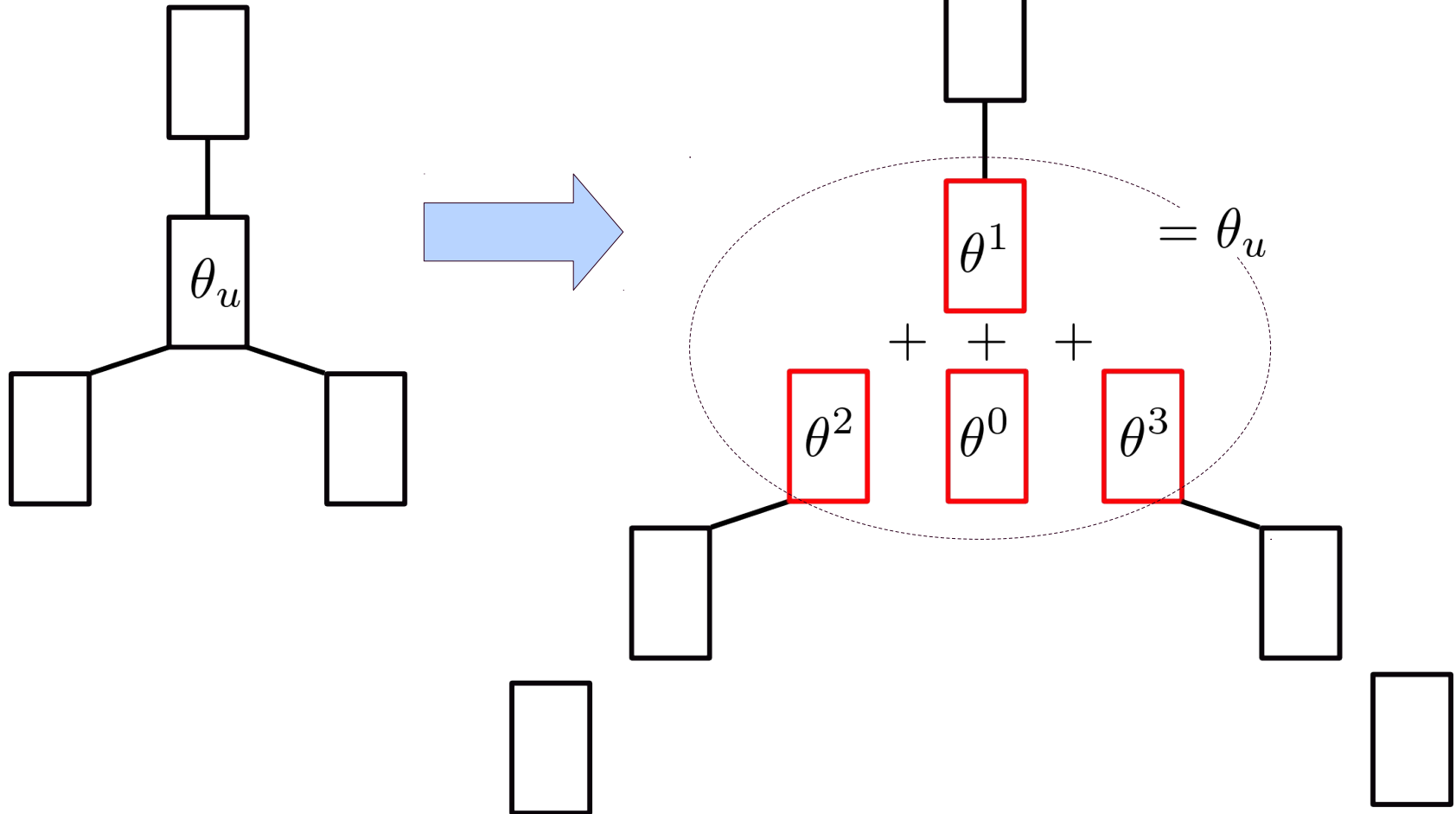


Exercise (easy): Show that

$$x^* = \arg \min_{x \in \mathcal{X}_v} E(x)$$

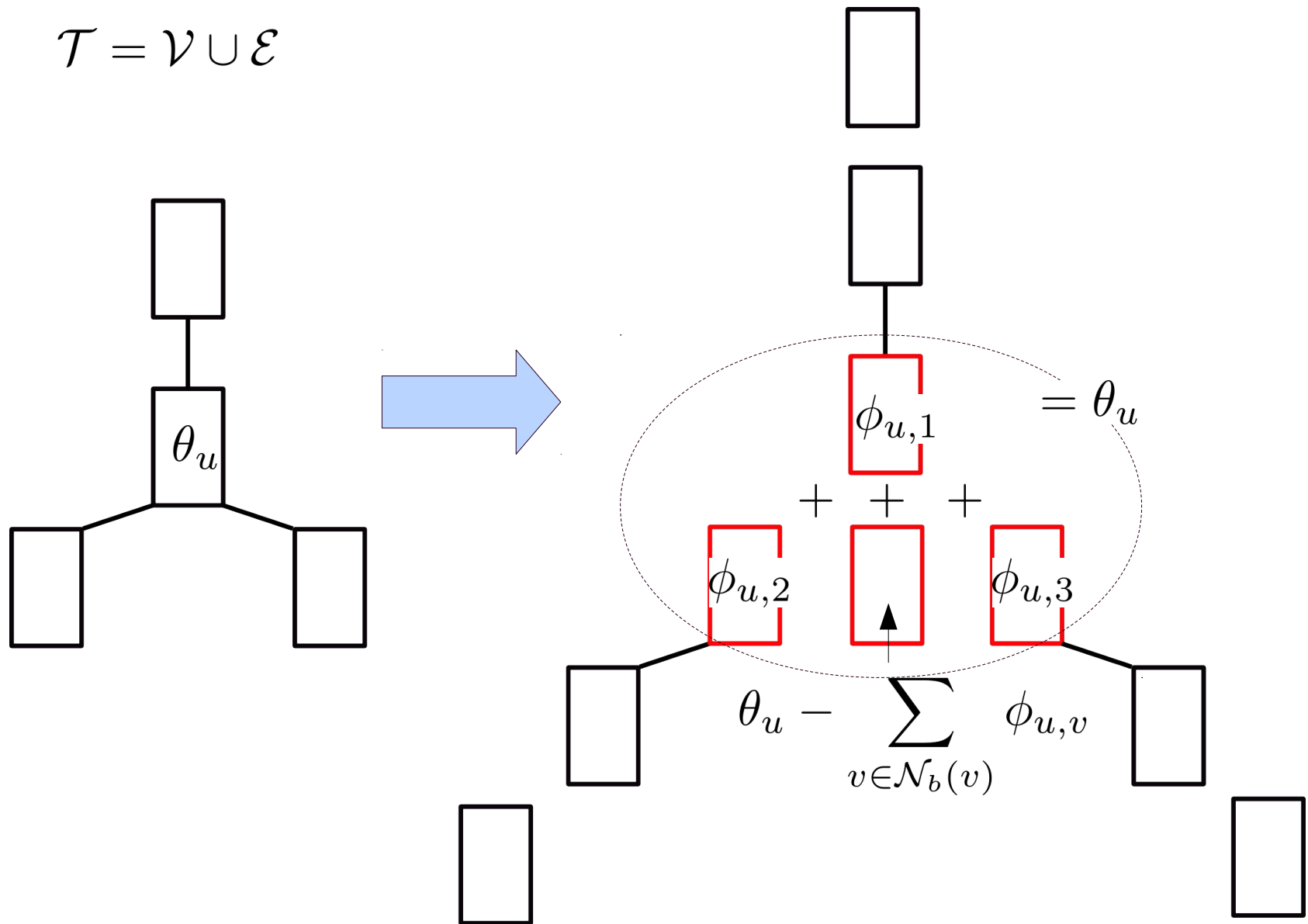
Complete Decomposition

$$\mathcal{T} = \mathcal{V} \cup \mathcal{E}$$



Complete Decomposition

$$\mathcal{T} = \mathcal{V} \cup \mathcal{E}$$



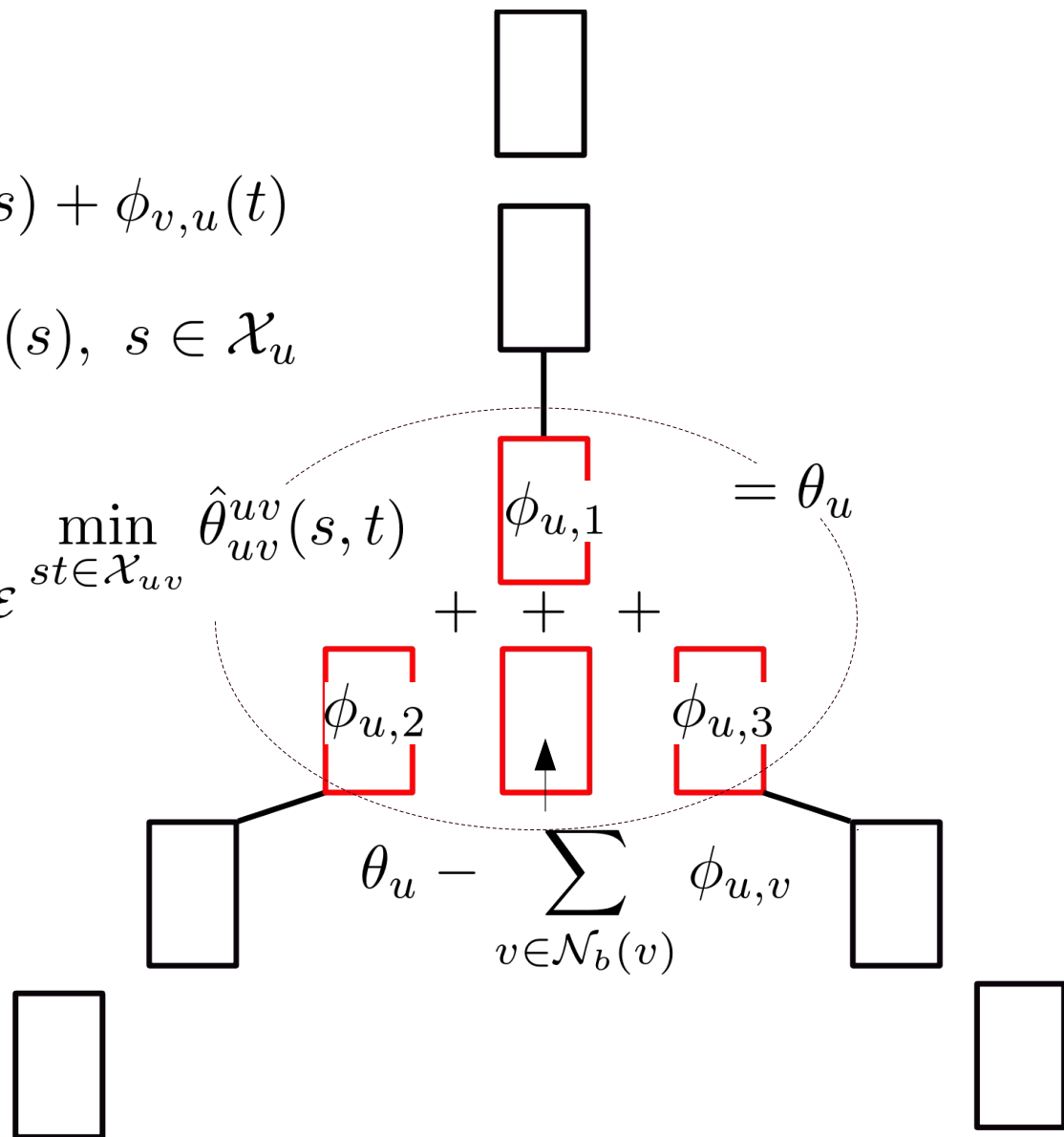
Complete Decomposition Dual = Lagrange Dual

$$\mathcal{T} = \mathcal{V} \cup \mathcal{E}$$

$$\hat{\theta}_{uv}^{uv}(s, t) = \theta_{uv}^{uv}(s, t) + \phi_{u,v}(s) + \phi_{v,u}(t)$$

$$\hat{\theta}_u^u(s) = \theta_u(s) - \sum_{v \in \mathcal{N}_b(u)} \phi_{u,v}(s), \quad s \in \mathcal{X}_u$$

$$U(\phi) = \sum_{u \in \mathcal{V}} \min_{s \in \mathcal{X}_u} \hat{\theta}_u^u(s) + \sum_{uv \in \mathcal{E}} \min_{st \in \mathcal{X}_{uv}} \hat{\theta}_{uv}^{uv}(s, t)$$



Complete Decomposition Dual = Lagrange Dual

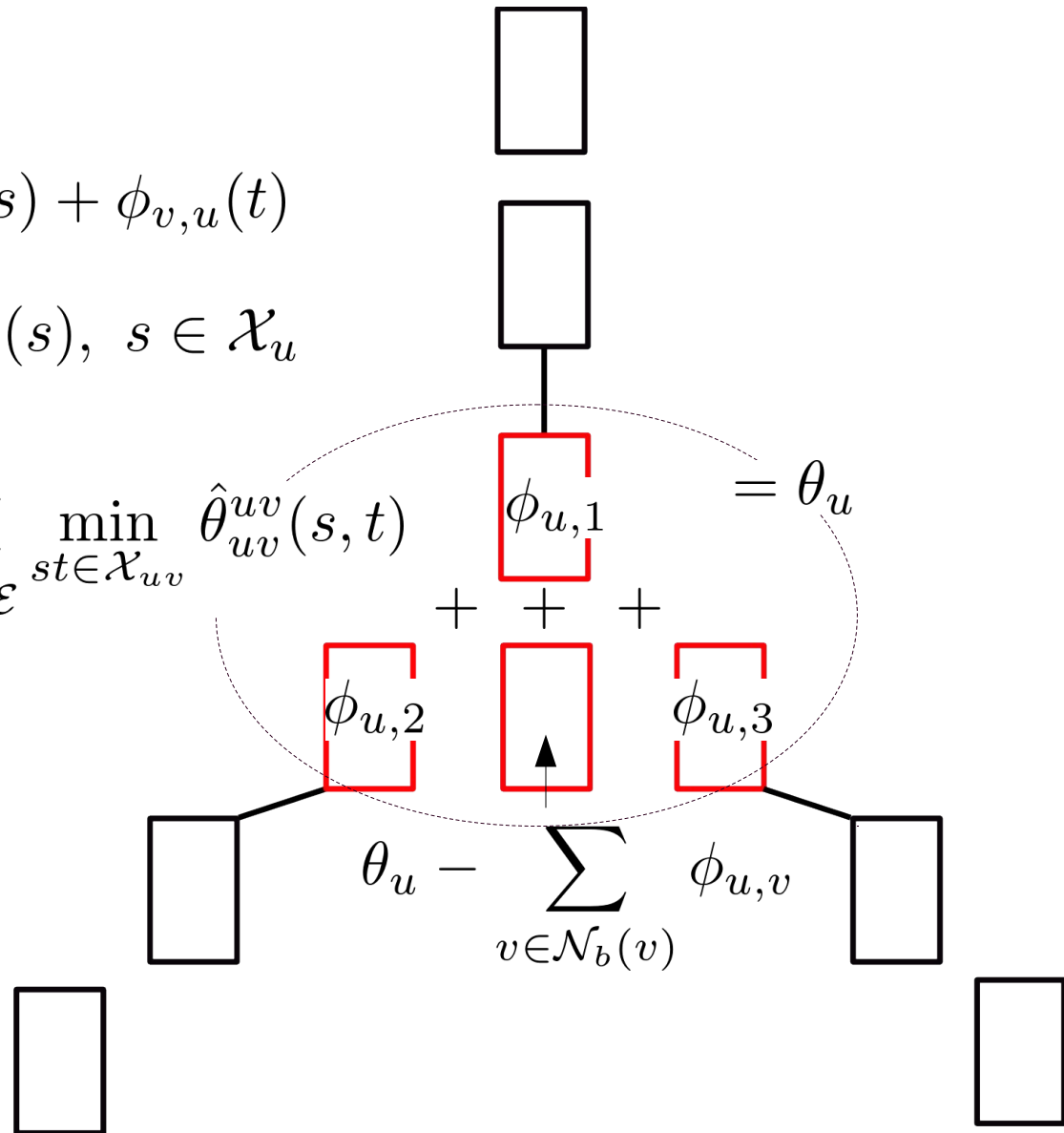
$$\mathcal{T} = \mathcal{V} \cup \mathcal{E}$$

$$\hat{\theta}_{uv}^{uv}(s, t) = \theta_{uv}^{uv}(s, t) + \phi_{u,v}(s) + \phi_{v,u}(t)$$

$$\hat{\theta}_u^u(s) = \theta_u(s) - \sum_{v \in \mathcal{N}_b(u)} \phi_{u,v}(s), \quad s \in \mathcal{X}_u$$

$$U(\phi) = \sum_{u \in \mathcal{V}} \min_{s \in \mathcal{X}_u} \hat{\theta}_u^u(s) + \sum_{uv \in \mathcal{E}} \min_{st \in \mathcal{X}_{uv}} \hat{\theta}_{uv}^{uv}(s, t)$$

$$D(\phi) = U(\phi)$$



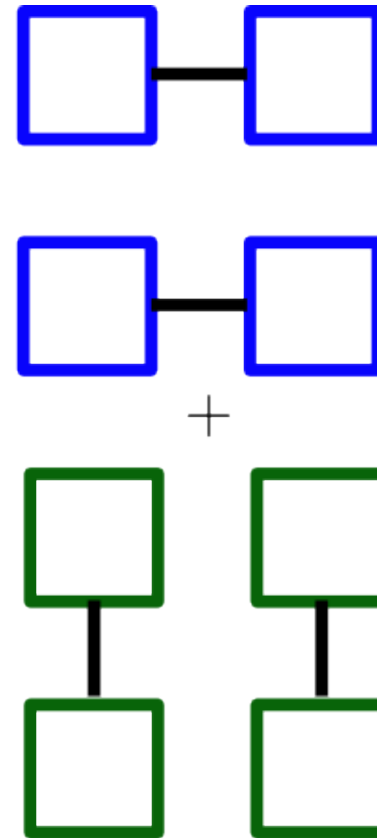
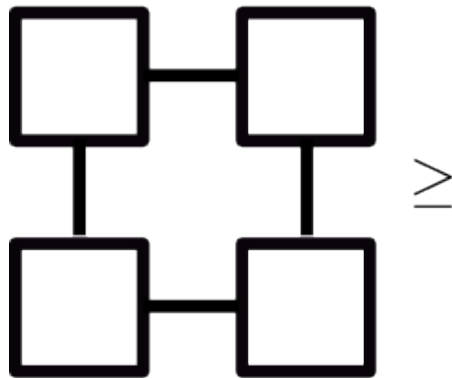
Implications:

Complete decomposition ~ Local polytope relaxation $\max_{\phi} U(\phi) = \min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle$

Complete decomposition is tight for **acyclic** graphs $\max_{\phi} U^{\tau}(\phi) = \min_{\mu \in \mathcal{M}^{\tau}} \langle \theta^{\tau}, \mu \rangle$

All tree decompositions are equivalent and correspond to the local polytope relaxation

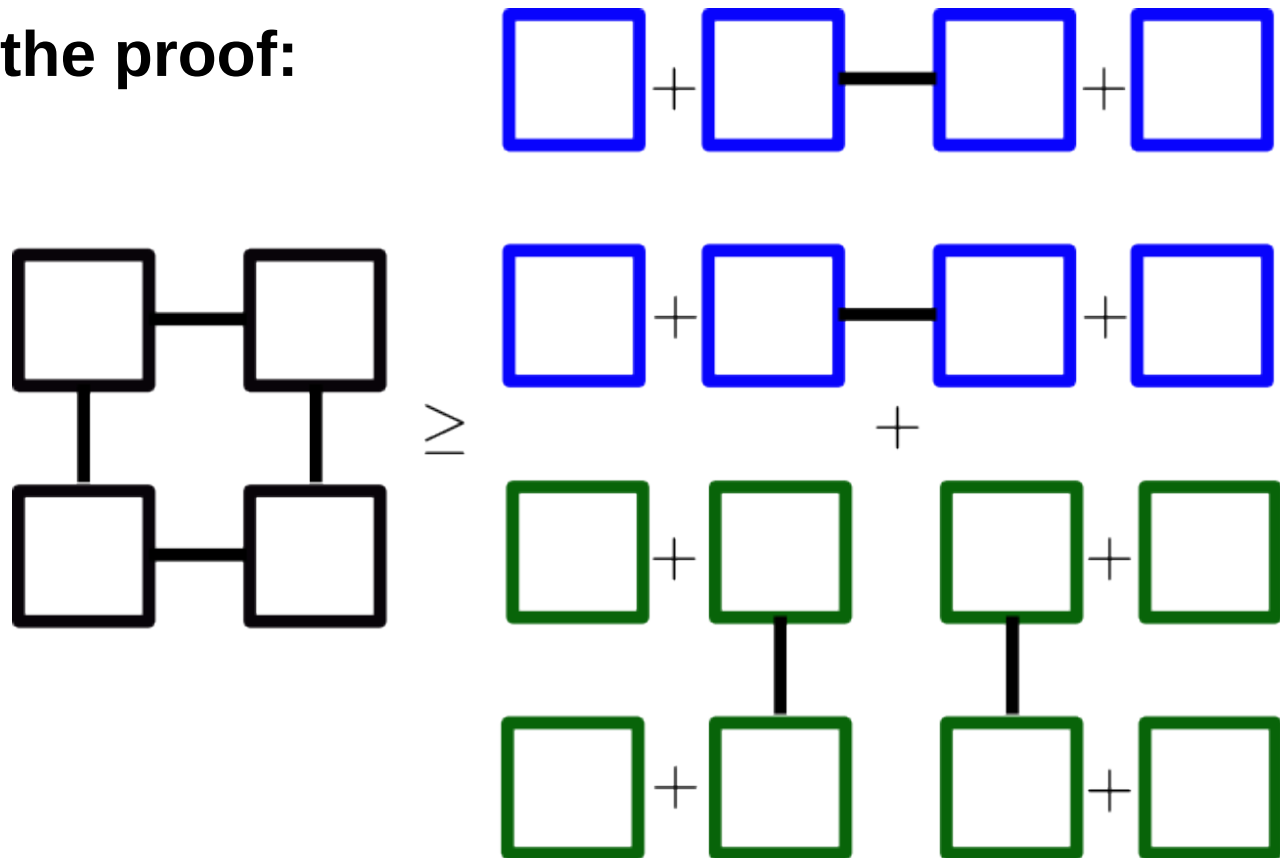
Sketch of proof:



Equivalence of Tree Decompositions

All tree decompositions are equivalent and correspond to the local polytope relaxation

Sketch of the proof:

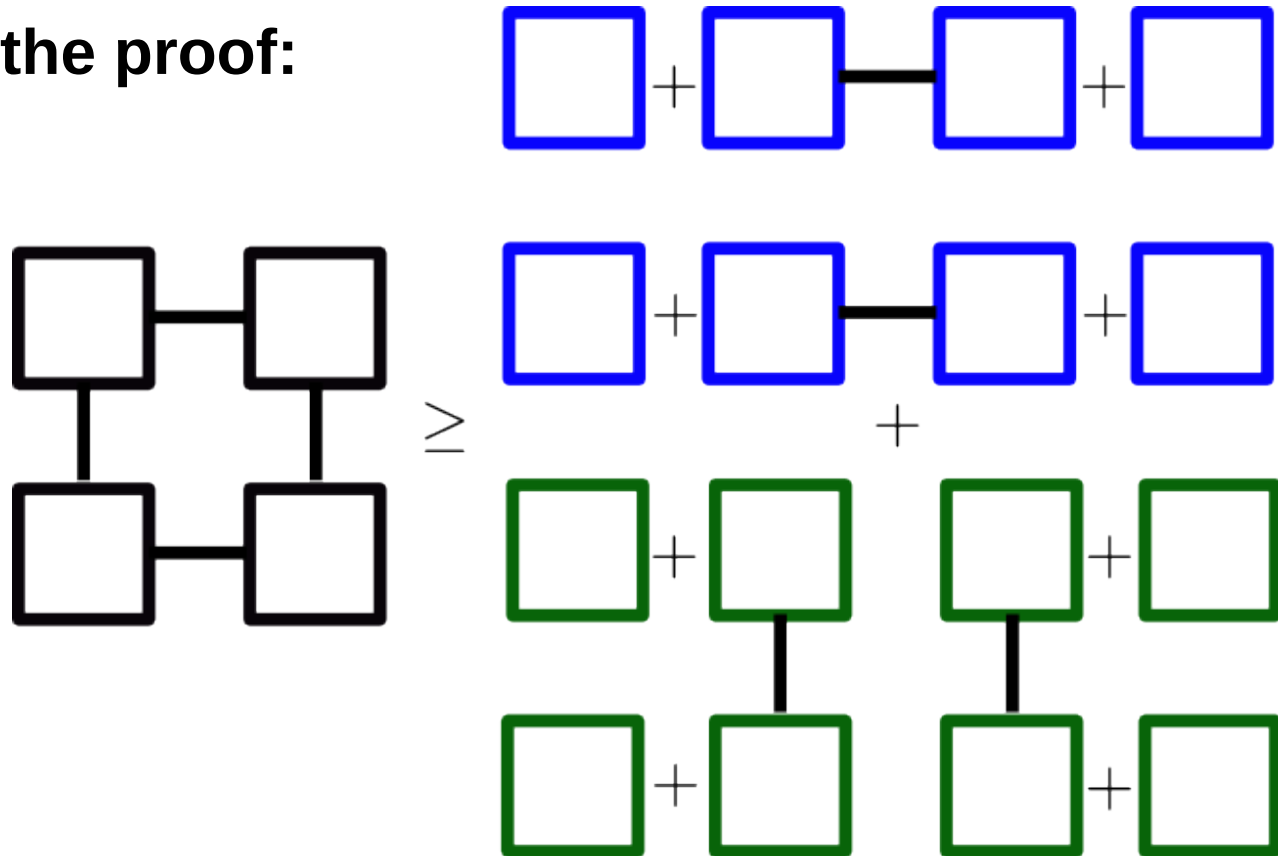


Equivalence of Tree Decompositions

Duplicate subproblems can be merged:

$$\min_x f(x) = \min_{x_1} \alpha f(x_1) + \min_{x_2} (1 - \alpha) f(x_2)$$

Sketch of the proof:



Lagrangian (dual) decomposition or Lagrangian relaxation:

- 1) Decompose to solvable sub-problems $\Rightarrow$ concave dual
- 2) If integer solutions are consistent $\Rightarrow$ non-relaxed energy minimum, otherwise - rounding
- 3) Dual optimum $\Rightarrow$ primal relaxed problem, relaxation depends on decomposition
- 4) Complete decomposition = dual to the local polytope relaxation
- 5) Tree decomposition $\Rightarrow$ local polytope relaxation

General (simplest) solution method: subgradient method

For complete decomposition: block-coordinate ascent (diffusion)

What about the coordinate ascent for a tree decomposition?